Nathanael Bosch

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Research Interests

Probabilistic machine learning and scientific simulation: surrogate models and learned simulators, neural operators, electronic structure simulations, density functional theory, state-space models, differential equations, Bayesian filtering and smoothing, probabilistic numerics.

Professional Experience

École polytechnique fédérale de Lausanne (EPFL) — **Lausanne, Switzerland**
Postdoctoral Researcher — *2025/10 – today*
Supervised by Michael Herbst, in the Mathematics for Materials Modelling group.

Amazon Web Services — **Berlin, Germany**
Applied Scientist Intern — *2024/08 – 2025/01*
Supervised by Oleksandr Shchur and Caner Türkmen, in the Machine Learning and Forecasting team of AWS AI Research.

University of Tübingen — **Tübingen, Germany**
Research Assistant — *2020/05 – 2024/06*
Supervised by Philipp Hennig, in the Methods of Machine Learning group.

Max Planck Institute for Intelligent Systems — **Tübingen, Germany**
Student Assistant — *2019/03 – 2019/10*
Supervised by Jörg Stückler, in the Embodied Vision Group.

Horváth&Partners — **Munich, Germany**
Data Science Intern — *2016/05 – 2016/07*

Education

University of Tübingen — **Tübingen, Germany**
Ph.D. Computer Science, Magna Cum Laude (with great distinction) — *2020/05 – 2025/02*
With the *International Max Planck Research School for Intelligent Systems (IMPRS-IS)*.
Advisor: Philipp Hennig.
Doctoral thesis title: "A Flexible and Efficient Framework for Probabilistic Numerical Simulation and Inference".

Technical University of Munich — **Munich, Germany**
M.Sc. Data Engineering and Analytics, German grade 1.3 (Distinction) — *2018/04 – 2019/10*

Technical University of Munich — **Munich, Germany**
M.Sc. Mathematics, German grade 1.2 (High Distinction) — *2016/10 – 2018/10*

Technical University of Munich — **Munich, Germany**
B.Sc. Mathematics, German grade 1.8 (Distinction) — *2012/10 – 2016/06*

Landesgymnasium für Hochbegabte — **Schwäbisch Gmünd, Germany**
Abitur, German grade 1.4 — *2007/09 – 2012/06*

Publications

NB. *"Stable and Scalable Probabilistic Numerical Solvers for Stiff and High-Dimensional ODEs"*. Conference on Neural Information Processing Systems, **NeurIPS**, 2026.

NB*, Niklas Frederik Schmitz*, and Michael F. Herbst. *"Euclidean Fourier Neural Operators"*. Representations for the Physical Sciences Workshop at NeurIPS, 2026.

Joanna Sliwa, Frank Schneider, NB, Agustinus Kristiadi, and Philipp Hennig. *"Low-Rank Filtering & Smoothing for Sequential Deep Learning"*. Transactions on Machine Learning Research, **TMLR**, 2026.

NB, Oleksandr Shchur, Nick Erickson, Michael Bohlke-Schneider, and Ali Caner Turkmen. *"Multi-layer Stack Ensembles for Time Series Forecasting"*. International Conference on Automated Machine Learning, **AutoML**, 2025.

Ori Press et al. *"AlgoTune: Can Language Models Speed Up General-Purpose Numerical Programs?"* Conference on Neural Information Processing Systems, **NeurIPS**, 2025.

Dingling Yao, Filip Tronarp, and NB. *"Propagating Model Uncertainty through Filtering-based Probabilistic Numerical ODE Solvers"*. International Conference on Probabilistic Numerics, **ProbNum**, 2025.

Jonas Beck, NB, Michael Deistler, Kyra L. Kadhim, Jakob H. Macke, Philipp Hennig, and Philipp Berens. *"Diffusion Tempering Improves Parameter Estimation with Probabilistic Integrators for Ordinary Differential Equations"*. International Conference on Machine Learning, **ICML**, 2024.

Amon Lahr, Filip Tronarp, NB, Jonathan Schmidt, Philipp Hennig, and Melanie N. Zeilinger. *"Probabilistic ODE Solvers for Integration Error-aware Numerical Optimal Control"*. Learning for Dynamics & Control Conference, **L4DC**, 2024.

NB. *"ProbNumDiffEq.jl: Probabilistic Numerical Solvers for Ordinary Differential Equations in Julia"*. Journal of Open Source Software, **JOSS**, 2024.

NB, Adrien Corenflos, Fatemeh Yaghoobi, Filip Tronarp, Philipp Hennig, and Simo Särkkä. *"Parallel-in-Time Probabilistic Numerical ODE Solvers"*. Journal of Machine Learning Research, **JMLR**, 2024.

NB, Philipp Hennig, and Filip Tronarp. *"Probabilistic Exponential Integrators"*. Conference on Neural Information Processing Systems, **NeurIPS**, 2023.

Nicholas Krämer*, NB*, Jonathan Schmidt*, and Philipp Hennig. *"Probabilistic ODE Solutions in Millions of Dimensions"*. International Conference on Machine Learning, **ICML**, 2022.

NB, Filip Tronarp, and Philipp Hennig. *"Pick-and-Mix Information Operators for Probabilistic ODE Solvers"*. International Conference on Artificial Intelligence and Statistics, **AISTATS**, 2022.

Filip Tronarp*, NB*, and Philipp Hennig. *"Fenrir: Physics-Enhanced Regression for Initial Value Problems"*. International Conference on Machine Learning, **ICML**, 2022.

NB, Philipp Hennig, and Filip Tronarp. *"Calibrated Adaptive Probabilistic ODE Solvers"*. International Conference on Artificial Intelligence and Statistics, **AISTATS**, 2021.

NB*, Jan Achterhold*, Laura Leal-Taixé, and Jörg Stückler. *"Planning from Images with Deep Latent Gaussian Process Dynamics"*. Learning for Dynamics & Control Conference, **L4DC**, 2020.

Preprints

Jonathan Wenger, Nicholas Krämer, Marvin Pförtner, Jonathan Schmidt, NB, Nina Effenberger, Johannes Zenn, Alexandra Gessner, Toni Karvonen, François-Xavier Briol, Maren Mahsereci, and Philipp Hennig. *"ProbNum: Probabilistic Numerics in Python"*. arXiv, 2021.

* indicates equal contribution.

Invited talks

Probabilistic Numerics Workshop (ProbNum24) **London, United Kingdom**
Talk: "Robust Parameter Inference in ODEs via Physics-Enhanced Gaussian Process Regression" *2024/07*

Probabilistic Numerics Spring School **Southampton, United Kingdom**
Practical session: "Probabilistic Numerics for ODEs" *2024/04*

International Conference on Scientific Computing and Machine Learning **Kyoto, Japan**
Talk: "Probabilistic Numerical Solvers for Differential Equations" *2024/03*

SIAM Conference on Uncertainty Quantification **Trieste, Italy**
Talk: "Probabilistic Numerics for Ordinary Differential Equations" *2024/02*

Statistics PhD seminar, University of Edinburgh — **Edinburgh, United Kingdom (virtual)**
Talk: "Probabilistic Numerics for Ordinary Differential Equations" — *2023/05*

Probabilistic Numerics Spring School — **Tübingen, Germany**
Practical session: "Probabilistic Numerics for ODEs" — *2023/03*

Institute for Dynamic Systems and Control, ETH Zurich — **Zurich, Switzerland**
Talk: "ODE Solvers as Gauss–Markov Regression" — *2023/01*

Sensor informatics and medical technology group, Aalto University — **Aalto, Finnland**
Talk: "Solving Differential Equations with Bayesian Filtering and Smoothing" — *2022/05*

Dagstuhl Seminar on Probabilistic Numerical Methods — **Wadern, Germany**
Talk: "ProbNumDiffEq.jl: Fast and Practical ODE Filters in Julia" — *2021/10*

Teaching Experience

EPFL — **Lausanne, Switzerland**
Teaching assistant: "Analysis I" — *2025 / 2026*
Course taught by Nicolas Boumal.

EPFL — **Lausanne, Switzerland**
Teaching assistant: "Linear Algebra I" — *2025 / 2026*
Course taught by Philippe Michel.

University of Tübingen — **Tübingen, Germany**
Teaching assistant / Co-lecturer: "Probabilistic Machine Learning" — *2023*
Course mainly taught by Philipp Hennig. In addition to the regular teaching assistant duties, I gave one full M.Sc.-level lecture on "Hidden Markov Models".

University of Tübingen — **Tübingen, Germany**
Co-lecturer: "Numerics of Machine Learning" — *2022 / 2023*
Gave two full M.Sc.-level lectures on "Ordinary Differential Equations" and "Probabilistic Numerical ODE Solvers", as part of a course taught by Philipp Hennig and multiple other PhD students from the Methods of Machine Learning group. This also included preparing and grading homework assignments.

Uppsala University — **Uppsala, Sweden**
Guest lecturer: "Probabilistic Numerics for Ordinary Differential Equations" — *2022*
Gave a single guest lecture as a part of a seminar on "A computational introduction to stochastic differential equations", organized and taught by Zheng Zhao.

University of Tübingen — **Tübingen, Germany**
Teaching assistant: "Data Literacy" — *2021 / 2022*
Course taught by Philipp Hennig.

University of Tübingen — **Tübingen, Germany**
Seminar: "Machine learning for and with dynamical systems" — *2021*
Jointly organized with Nicholas Krämer and Philipp Hennig.

University of Tübingen — **Tübingen, Germany**
Teaching assistant: "Time Series" — *2020 / 2021*
Course taught by Filip Tronarp.

Technical University of Munich — **Munich, Germany**
Teaching assistant: "Principles of Mathematics 2" — *2017*
Mathematics course, aimed at B.Sc. Engineering Science students.

abiturma GbR — **Munich and Stuttgart, Germany**
Course instructor — *2016 & 2017*
Intensive five-day preparation course for the german Abitur in mathematics.

Supervision

Miguel Francisco Magro Dos Reis, Master's semester project — 2026
"Euclidean Fourier Neural Operators for Electronic Density Prediction".

Andrés Santiago Ulloa Canizares, Bachelor's student project — 2026
"Bayesian Optimization for Crystal Structure Prediction".

Fabrice van der Lehr, Master's thesis 2024
"Bayesian Filtering for Black Box Simulators"; co-supervised with Tobias Weber.

Sofiya Garkot, Essay rotation 2023 / 2024
"Exploring Probabilistic Solvers for Hodgkin-Huxley Models"; co-supervised with Jonas Beck.

Thomas Albrecht, Master's thesis 2021 / 2022
"Bayesian Physics-Informed Neural Networks via Laplace Approximations".

Felix Böhm, Bachelor's thesis 2021 / 2022
"Inferring ODE Parametric Latent Forces via Neural ODEs".

Dingling Yao, Master's thesis 2021
"Uncertainty Propagation in Probabilistic Ordinary Differential Equation Solvers"; co-supervised with Filip Tronarp.

Joanna Sliwa, Essay rotation 2021
"Physics-Informed Neural Networks"; co-supervised with Nicholas Krämer.

Academic services

Program committee for the *2nd International Conference on Probabilistic Numerics*. 2026

Review services

International Conference on Machine Learning (ICML)	2022*, 2023*, 2024
Conference on Neural Information Processing Systems (NeurIPS)	2021, 2022, 2023
International Conference on Artificial Intelligence and Statistics (AISTATS)	2021, 2022, 2023, 2024, 2026
Transactions on Machine Learning Research (TMLR)	2022 – 2023
International Conference on Probabilistic Numerics (ProbNum)	2025, 2026

* indicates *outstanding reviewer* awards.

Software

`ProbNumDiffEq.jl`
Maintainer
Probabilistic Numerical Differential Equation solvers via Bayesian filtering and smoothing in Julia; compatible with the popular DifferentialEquations.jl / SciML ecosystem. Package presented at the JuliaCon 2024 in Eindhoven.

Technical skills

Working knowledge: Julia, Python, jax, NumPy/SciPy, PyTorch, git/GitHub, LaTeX, Unix, GNU Emacs.

Basic knowledge: C/C++, SLURM, MATLAB, R, SQL.

Language skills

German: Native speaker

French: Near native

English: Fluent

Spanish: Good working knowledge

Russian: Basic communication skills

Scholarships & Awards

Seal of Excellence – Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship 2026
Awarded the European Commission's Seal of Excellence for an MSCA Postdoctoral Fellowship proposal.

Leadership Talent Academy 2025
Selected participant in the University of Tübingen's Leadership Talent Academy, a 3-month program of intensive weekend modules and coaching sessions focused on developing leadership skills and personal growth.

German Academic Scholarship Foundation 2013 – 2019

Recipient of the German Academic Scholarship Foundation, providing financial and academic support to selected students.

Siemens Mentoring Program 2017 – 2018

Selected participant in the Siemens Mentoring Program, pairing students with Siemens professionals for career development and mentorship.